Milestone 4 - System Prototype

Team GitHub notebook URL: https://github.com/Kongbai815/job-matching-nlp/blob/main/MSBA_Job_Matching_Milestone4_System_Prototype.ipynb

1. End-to-End Architecture

The system separates search requests from posting classification, enforces verifiable search constraints, scores postings with a trained word+character TF-IDF SVM, applies calibrated uncertainty and deterministic governance gates, and supports both individual and batch review.

Stage	Implementation	Output
Route	search vs. posting input mode	Correct task semantics
Retrieve	80k sparse index + hard constraints + dedupe	Top-6 evidence
Classify	LinearSVC trained on 77,135 deduplicated rows	Four-class role fit
Govern	calibrated margin + deterministic gates	Review / Hold
Generate	deterministic evidence template	Citations and caveat

2. Executed Setup

```
from msba_job_matcher.core import JobMatchingSystem
system = JobMatchingSystem(DATA, model_path=MODEL)
```

```
Project rows: 100000
Retrieval rows: 80000
Classifier: word+character TF-IDF LinearSVC (77,135 deduplicated training rows)
```

3. Workflow Inputs

Case	Mode	Model label	System label	Review
entry level analytics search	search	N/A (search request)	N/A (search request)	True
senior engineering posting	posting	low_fit	low_fit	True
authorization restriction posting	posting	high_fit	high_fit	True
thin metadata posting	posting	medium_fit	unclear	True

4. Retrieved Evidence and Policy Behavior

Case	Top evidence	Country	Authorization / policy
entry level analytics search	Data Analyst	United States	not available in public source
senior engineering posting	Data Engineer	United States	not available in public source
authorization restriction posting	Jr. Data Analyst (No OPT/CPT)	United States	explicit source text: no OPT/CPT
thin metadata posting	Summer Data Analytics Internship	United States	Missing: company, location, country, schedule

5. Evaluation Against Baseline

System	Rows	Accuracy	Macro F1
M2 same-subset baseline	4,000	-	0.791
M3 true PEFT LoRA	4,000	0.875	0.873
M4 trained classifier	4,000	0.984	0.984

Same-subset macro-F1 improvement: **+0.192**. All four final per-label F1 scores are at least 0.970.

6. Leakage-Controlled Checks

Validation slice	Rows	Macro F1
Fixed set, novel text	3,751	0.983
Full validation, novel text	18,672	0.979
Full validation, unseen companies	5,570	0.980
Newest 20% by posting date	20,000	0.976

Selective review audit: **91.6% coverage** at **99.47% weak-label accuracy** on a disjoint 10,000-row audit. Search audit: 36 results, 0 hard-constraint violations, 0 duplicate company-title pairs.

7. Grounding, Hallucination, and Governance

All four cases returned retrieved evidence. The authorization case demonstrates the critical conflict: the classifier predicts high role fit, but explicit No OPT/CPT text forces a Hold and advisor review. The thin-metadata case demonstrates a second policy override: missing company, location, country, and schedule fields route the result to unclear. The saved casebook contains zero unsupported authorization claims.

8. What the Model Comparison Means

A controlled learning curve rises from 0.898 macro F1 at 2,000 rows to 0.984 at 80,000 rows. The final linear classifier also exceeds the 2,000-row LoRA result while producing a 3.8 MB artifact and training in under one minute. The time-based test confirms that the result is not limited to a random split.

9. Completed Validation Assets

The project scanned all 785,741 source rows and found 230 unique title-level authorization candidates. It includes a 430-row candidate/control benchmark, a 1,000-row advisor queue, a calibrated review policy, a six-query search audit, and a three-input batch example. Human fields remain blank so no model output is presented as advisor ground truth.

10. Reading Connection

HOLLM Chapter 8 motivates retrieval as external grounding. Tunstall Chapter 7 and Jurafsky & Martin Chapter 11 emphasize supported QA and extraction. The prototype follows that logic by keeping search, posting classification, retrieved evidence, missing-data policy, and authorization evidence as separate inspectable stages.